

Neural Network Task Division

1. Saif — Dataset & Code Comments

- Create and finalize the `student_dataset.csv` file (Study Hours, Assignments, Classes Missed → Pass/Fail).
- Write the Python code to load the dataset.
- Add clear, easy-to-understand comments to the final combined group code.

2. Farzana — Forward Pass Math & Backpropagation

- Write the `forward_pass` function ($z = X \cdot W + b$, Sigmoid activation).
- Ensure the terminal prints inputs, weights, bias, summation, and output.
- Implement one simple backprop step using the shortcut: $dL/dz = a - y$, then $dW = (a - y) \cdot X$, $db = (a - y)$.
- Apply a single weight update: $W_{new} = W - lr \cdot dW$.
- Hand-calculate one full forward + backward pass by hand for one sample (show old W vs. new W).

3. Isbat — Visualization & Diagrams

- Build the forward pass diagram (input circles → weights → summation → Sigmoid → output).
- Add a simple "before vs. after" bar chart comparing weights before and after the one backprop update.
- Generate the final `visualize.py` code to draw both.

4. Erin — Integration & Testing

- Combine dataset, forward math, backprop step, and diagrams into one `main.py`.
- Run: forward pass → compute loss → one backprop update → forward pass again, showing the output moved closer to the true label.
- Test this full cycle on 3 different student inputs.

5. Fatima — Full Presentation

- Build the entire slide deck: task overview, dataset, forward pass math, diagrams, the simple backprop step, and results.
- Include a slide on why Sigmoid was chosen, and one clearly showing the forward $\rightarrow$ loss $\rightarrow$ backward $\rightarrow$ updated weight cycle.
- Coordinate rehearsal and speaking parts.